
ECSE 4660/6660: Internetworking of Things

Lecture 15: Network Layer – Part III:
ARP, Routing, IPv6

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Spring 2026

Administrative

- Use Piazza for any questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Lab
 - TA hours 4-5 PM daily in 5312 lab
 - Lab 3 due Mon, 3/9 (BLE, Zbee)
 - Lab 4 due Fri, 3/20 (DSR, AODV, ARP)
- HW
 - HW3 released, due Mon, 3/16
 - No Late submission allowed
- Remote class 2/26, ~~3/23~~ 3/30

- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Next Steps
 - Work with TA to get the hardware needed (respond on piazza)
 - Mid-term report due 3/20 (1 weeks)
 - In-class presentation+2min video 4/20, 4/23
 - Live-demo 4/27
 - Final code/report 5/1

Sources: Internet Protocol (IP)

- IPv4/v6 References:

- J. Kurose and K. Ross, Computer Networking: A Top-Down Approach Featuring the Internet, 7th edition, 2017.
 - Chapter 4: IPv4
- Behrouz A. Forouzan, TCP/IP Protocol Suite, 4th edition, 2009.
 - Chapters 5-8: IPv4 and ARP
 - Chapters 26-27: IPv6

Network layer: “data plane” roadmap

- Network layer: overview
- IP: the Internet Protocol
 - Datagram format and IP Addressing
 - DHCP
 - **ARP**
 - Routing Table Lookup
 - IPv6



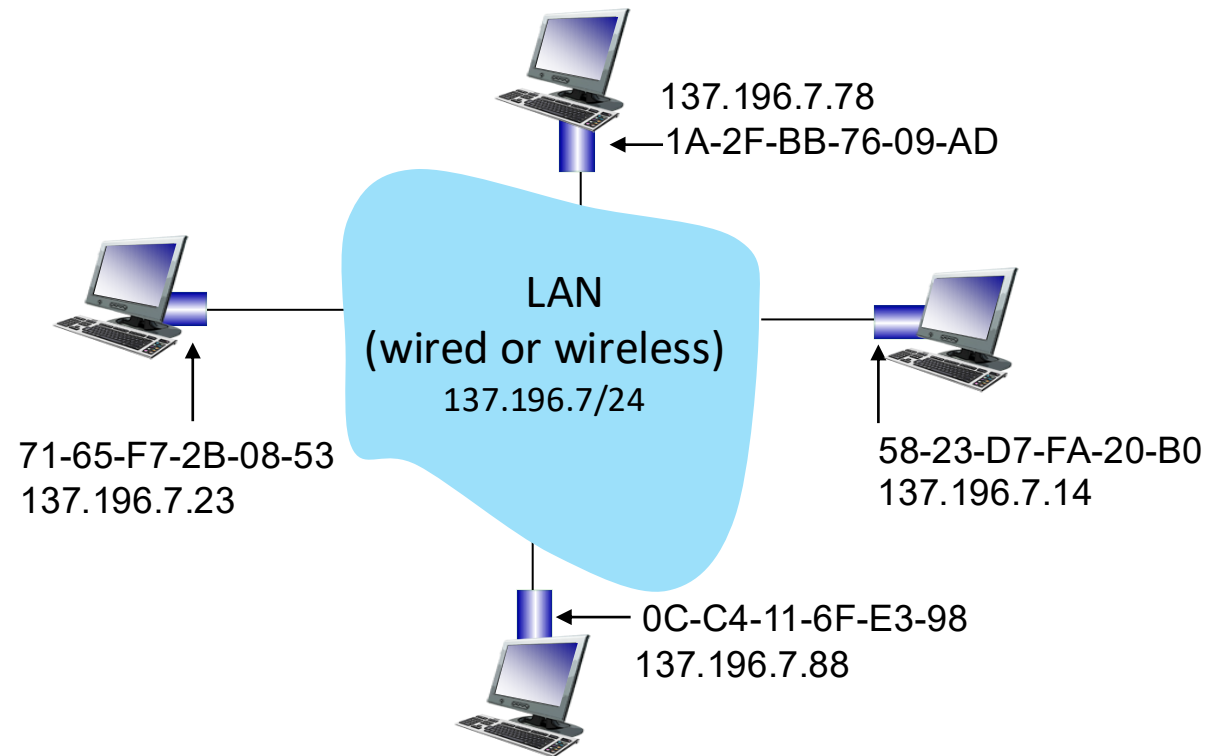
MAC (Media Access Control) addresses - Recap

- 32-bit IP address:
 - *network-layer* address for interface
 - used for layer 3 (network layer) forwarding
 - e.g.: 128.119.40.136
- MAC (or LAN or physical or Ethernet) address:
 - function: used “locally” to get frame from one interface to another physically-connected interface (same subnet, in IP-addressing sense)
 - 48-bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
 - e.g.: 1A-2F-BB-76-09-AD
 - hexadecimal (base 16) notation
(each “numeral” represents 4 bits)

MAC addresses

each interface on LAN

- has unique 48-bit **MAC** address
- has a locally unique 32-bit IP address (as we've seen)



MAC addresses

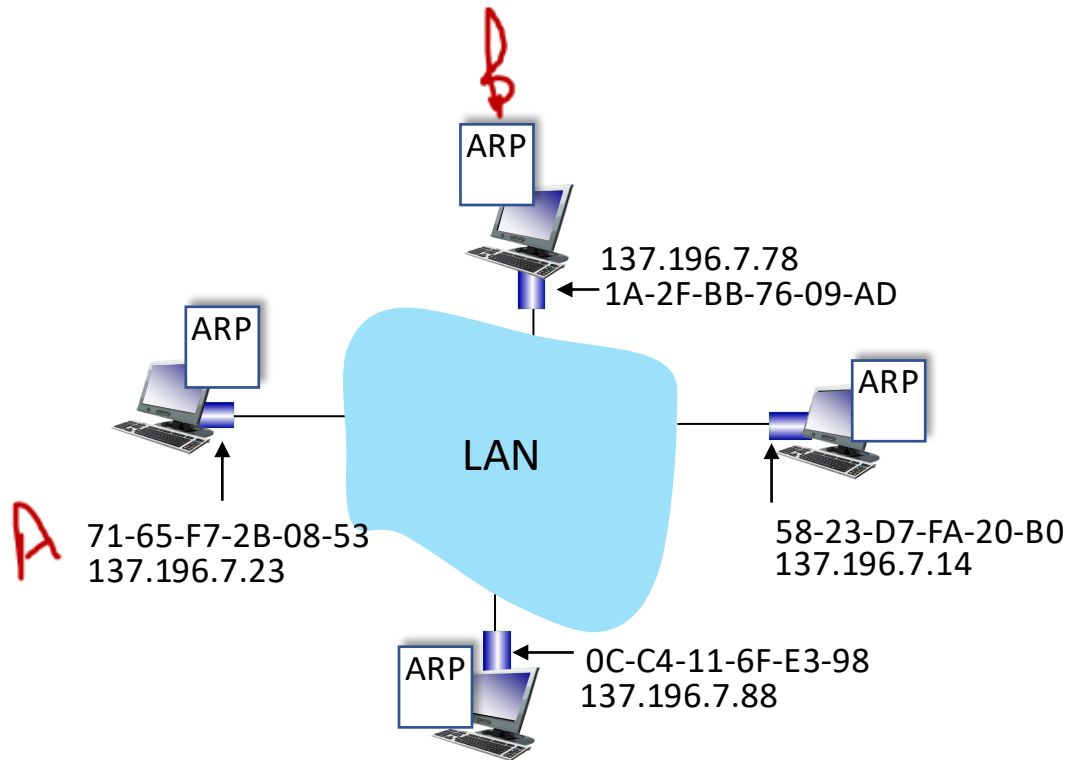
- MAC address allocation administered by IEEE
- manufacturer buys portion of MAC address space (to assure uniqueness)
- analogy:
 - MAC address: like Social Security Number
 - IP address: like postal address
- MAC flat address: portability
 - can move interface from one LAN to another
 - recall IP address *not* portable: depends on IP subnet to which node is attached

ARP: address resolution protocol

Question: how to determine interface's MAC address, knowing its IP address?

ARP table: each IP node (host, router) on LAN has table

- IP/MAC address mappings for some LAN nodes:
< IP address; MAC address; TTL >
- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)



ARP protocol in action

example: A wants to send datagram to B

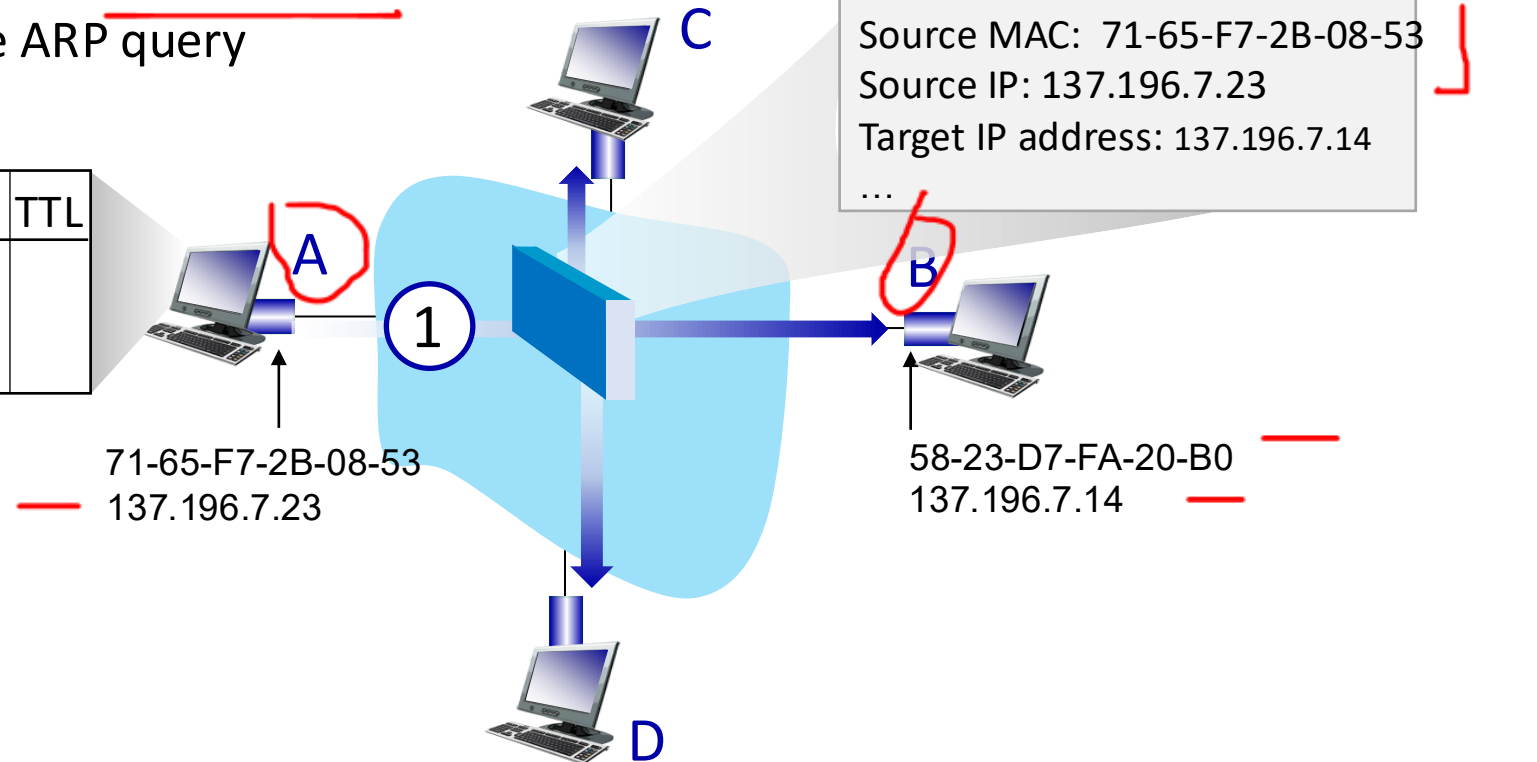
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address

A broadcasts ARP query, containing B's IP addr

- ①
- destination MAC address = FF-FF-FF-FF-FF-FF
 - all nodes on LAN receive ARP query

ARP table in A

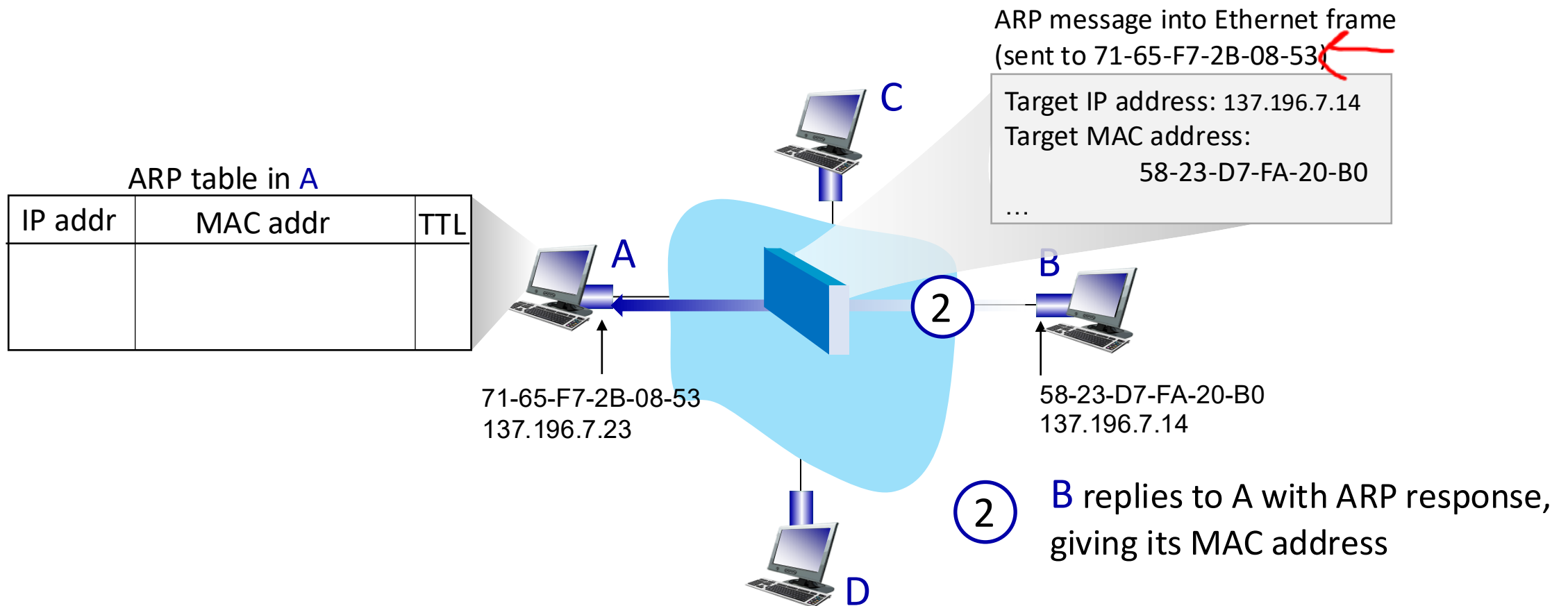
IP addr	MAC addr	TTL



ARP protocol in action

example: A wants to send datagram to B

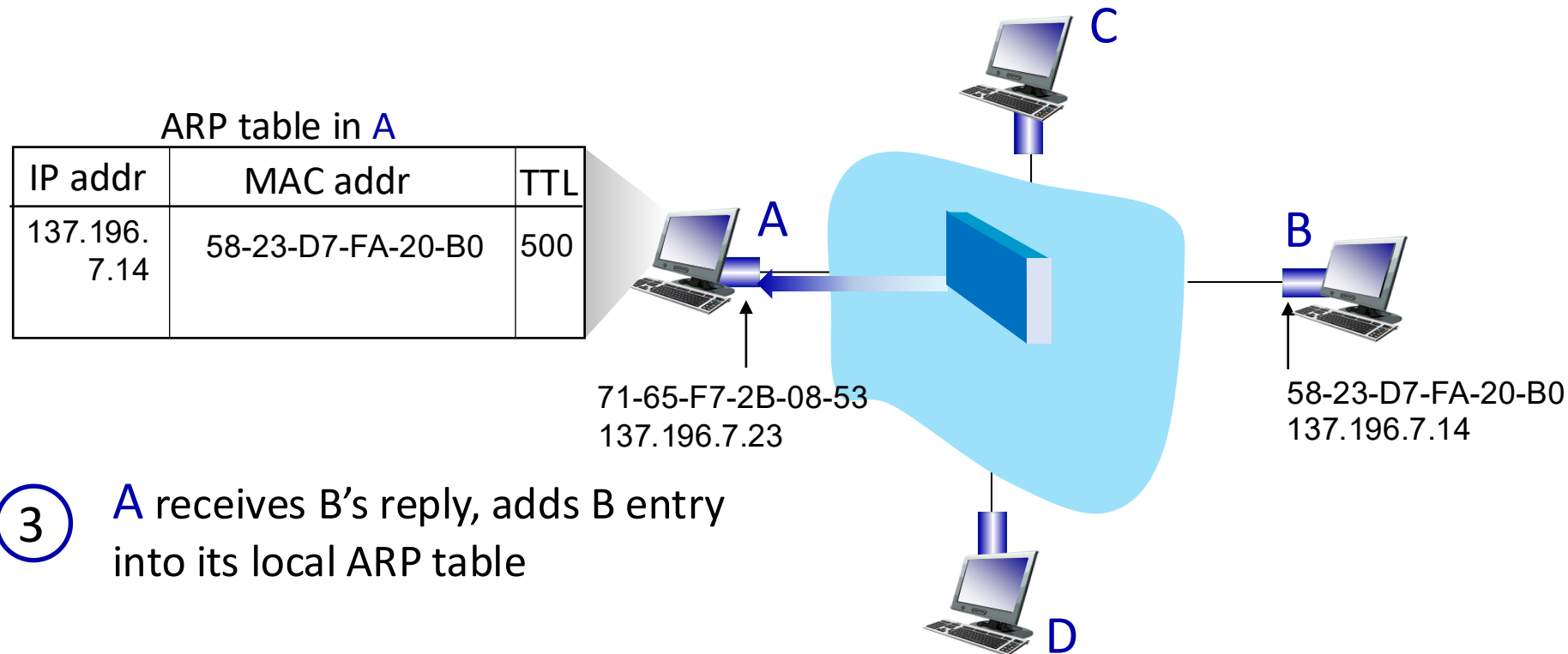
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



ARP protocol in action

example: A wants to send datagram to B

- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



ARP Packet Format

1	Hardware Type		Protocol Type
	Hardware length	Protocol length	Operation Request 1, Reply 2
2	Sender hardware address (For example, 6 bytes for Ethernet)		
4	Sender protocol address (For example, 4 bytes for IP)		
1	Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
4	Target protocol address (For example, 4 bytes for IP)		

ARP

ARP Packet Format (28 Byte)

Field	Size (Bytes)	Description
Hardware Type	2	Specifies the link-layer protocol type. (1 = Ethernet)
Protocol Type	2	Specifies the higher-layer protocol (0x0800 = IPv4).
Hardware Length	1	Length of the MAC address (6 for Ethernet).
Protocol Length	1	Length of the IP address (4 for IPv4).
Operation (Opcode)	2	Defines request (1) or reply (2).
Sender Hardware Address	6	MAC address of the sender (who is making the request).
Sender Protocol Address	4	IP address of the sender.
Target Hardware Address	6	MAC address of the target. In a request, this is <u>00:00:00:00:00:00</u> .
Target Protocol Address	4	IP address whose MAC address is being requested.

ARP Packet Format

```
▼ Frame 104: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface en0, id 0
  ► Interface id: 0 (en0)
    Encapsulation type: Ethernet (1)
    Arrival Time: Feb 26, 2020 18:43:06.784327000 EST
    [Time shift for this packet: 0.000000000 seconds]
    Epoch Time: 1582760586.784327000 seconds
    [Time delta from previous captured frame: 0.000209000 seconds]
    [Time delta from previous displayed frame: 0.000209000 seconds]
    [Time since reference or first frame: 2.048122000 seconds]
    Frame Number: 104
    Frame Length: 42 bytes (336 bits)
    Capture Length: 42 bytes (336 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:arp]
    [Coloring Rule Name: ARP]
    [Coloring Rule String: arp]
  ▼ Ethernet II, Src: XIAOMIEl_26:e5:0d (34:ce:00:26:e5:0d), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    ► Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    ► Source: XIAOMIEl_26:e5:0d (34:ce:00:26:e5:0d)
    Type: ARP (0x0806)
  ▼ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: XIAOMIEl_26:e5:0d (34:ce:00:26:e5:0d)
    Sender IP address: 192.168.31.1
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.31.200
```

Network layer: “data plane” roadmap

- Network layer: overview
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 - DHCP
 - ARP
 - Routing Table Lookup
 - IPv6



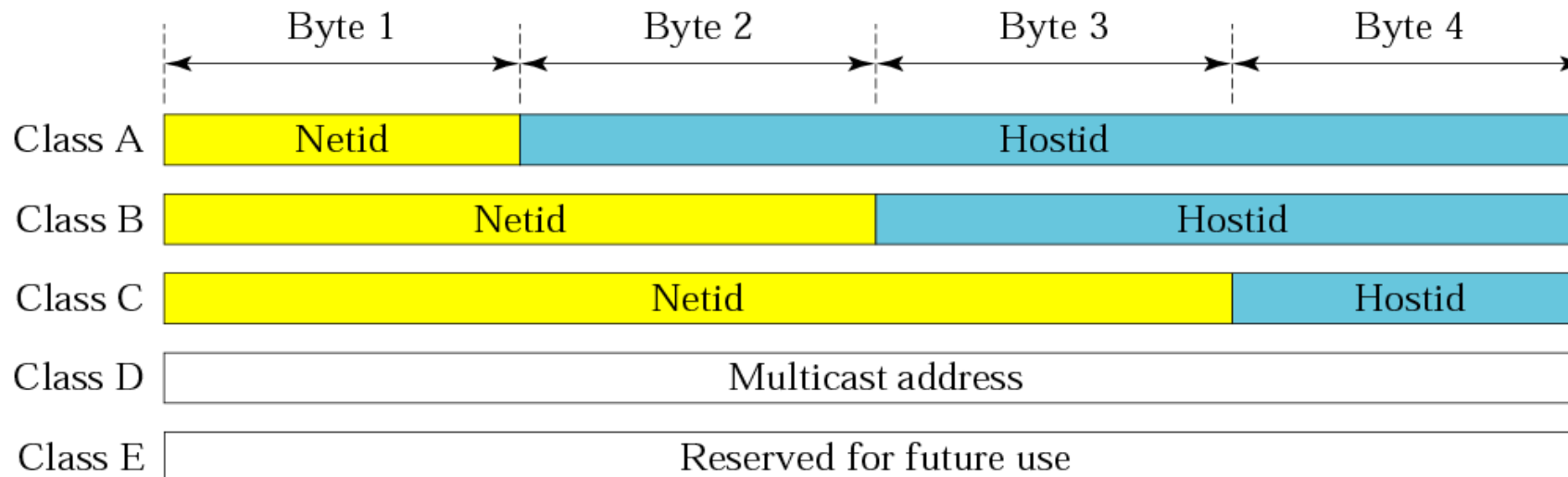
Classful Addressing

Five address classes:

	First byte	Second byte	Third byte	Fourth byte	
Class A	0				0.0.0.0 - 127.255.255.255
Class B	10				128.0.0.0 - 191.255.255.255
Class C	110				192.0.0.0 - 223.255.255.255
Class D	1110				224.0.0.0 - 239.255.255.255
Class E	1111				240.0.0.0 - 255.255.255.255

← 32 bits →

Large networks (e.g., big enterprises, governments)
Medium-sized organizations
Small-sized organizations

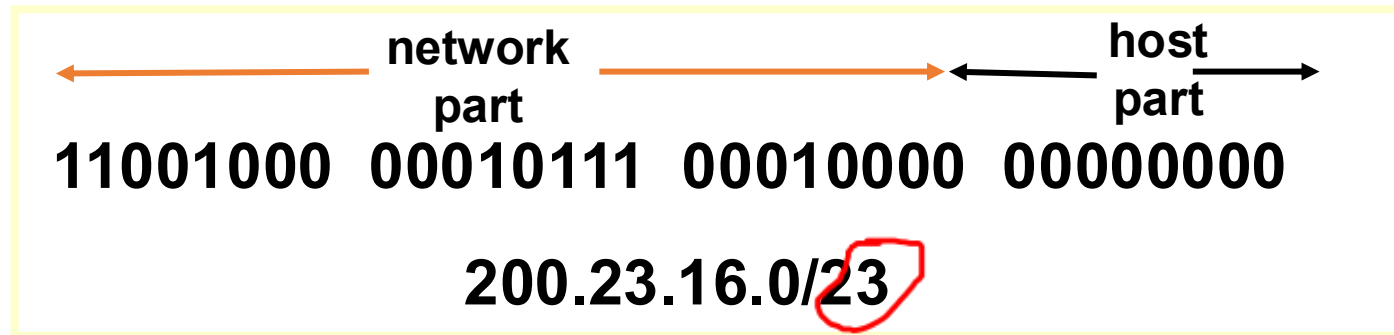


Classful vs. Classless Addressing

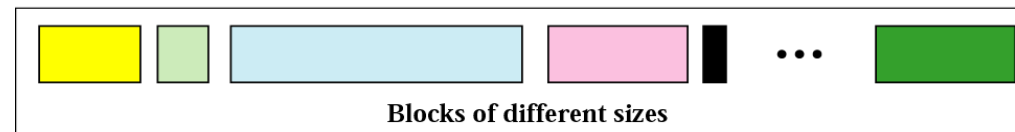
Limitation of classful addressing:

- inefficient use of address space, address space exhaustion (e.g., class B network allocated enough addresses for 65K hosts, even if only 2K hosts in that network)

- **CIDR: Classless InterDomain Routing**
network portion of address of arbitrary length
address format: a.b.c.d/x, x is # bits in network portion of address



Address Space

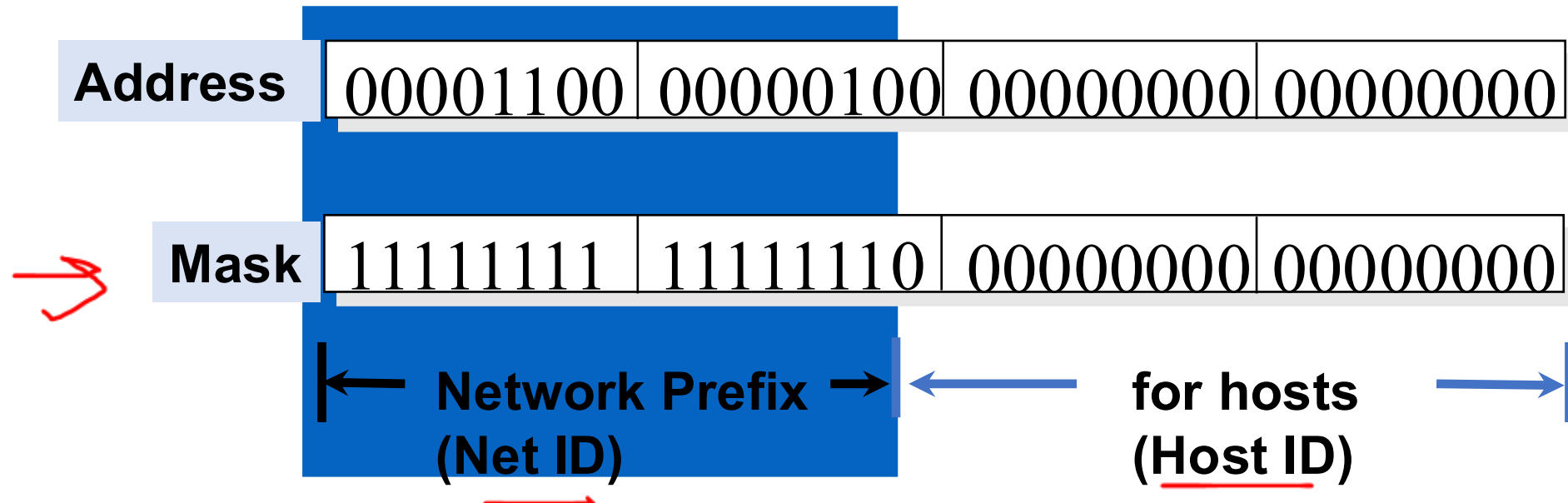


[src: Forouzan]

Network Address and Masks

Network mask: Put all 1s in the network part of the address, and all 0s in the host part

IP Address : 12.4.0.0/15 $\Leftrightarrow$
(Network address: 12.4.0.0, Mask: 255.254.0.0)



Understanding Prefixes and Masks

Does the IP 12.5.9.16 belong to this network? How about 12.7.9.16?

12.5.9.16 is covered by prefix 12.4.0.0/15 (It is part of this network)

12.5.9.16	00001100	00000101	00001001	00010000	
12.4.0.0/15	00001100	00000100	00000000	00000000	NetID
	11111111	11111110	00000000	00000000	Mask
12.7.9.16	00001100	00000111	00001001	00010000	

12.7.9.16 is not covered by prefix 12.4.0.0/15 (ie not part of that network)

Next-Hop Routing

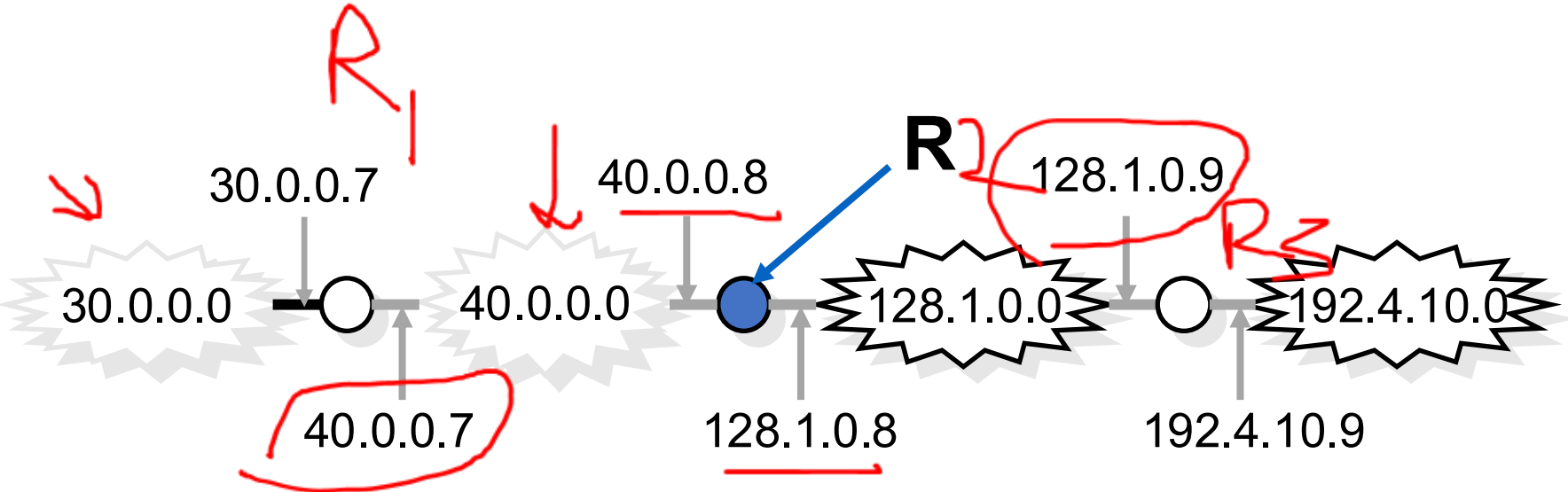
- **Direct delivery**: send datagram directly through Layer 2 (Ethernet, ...) when the source and the destination are on the **same (sub)network**.
- **Indirect delivery**: when the source and the destination are NOT on the same network
 - Need to send datagram through a router.
 - Consult the routing table to determine the next hop router.
 - Only ONE hop on the path is listed in the routing table.

Route Table Lookup (1/2)



- Mask must have contiguous 1s followed by contiguous zeros. Routers do not support other types of masks.
 - Route table lookup:
 - IF ((Mask[i] & Destination Addr) ==
Destination[i])
Forward to NextHop[i]
- 

Route Table Lookup (2/2)

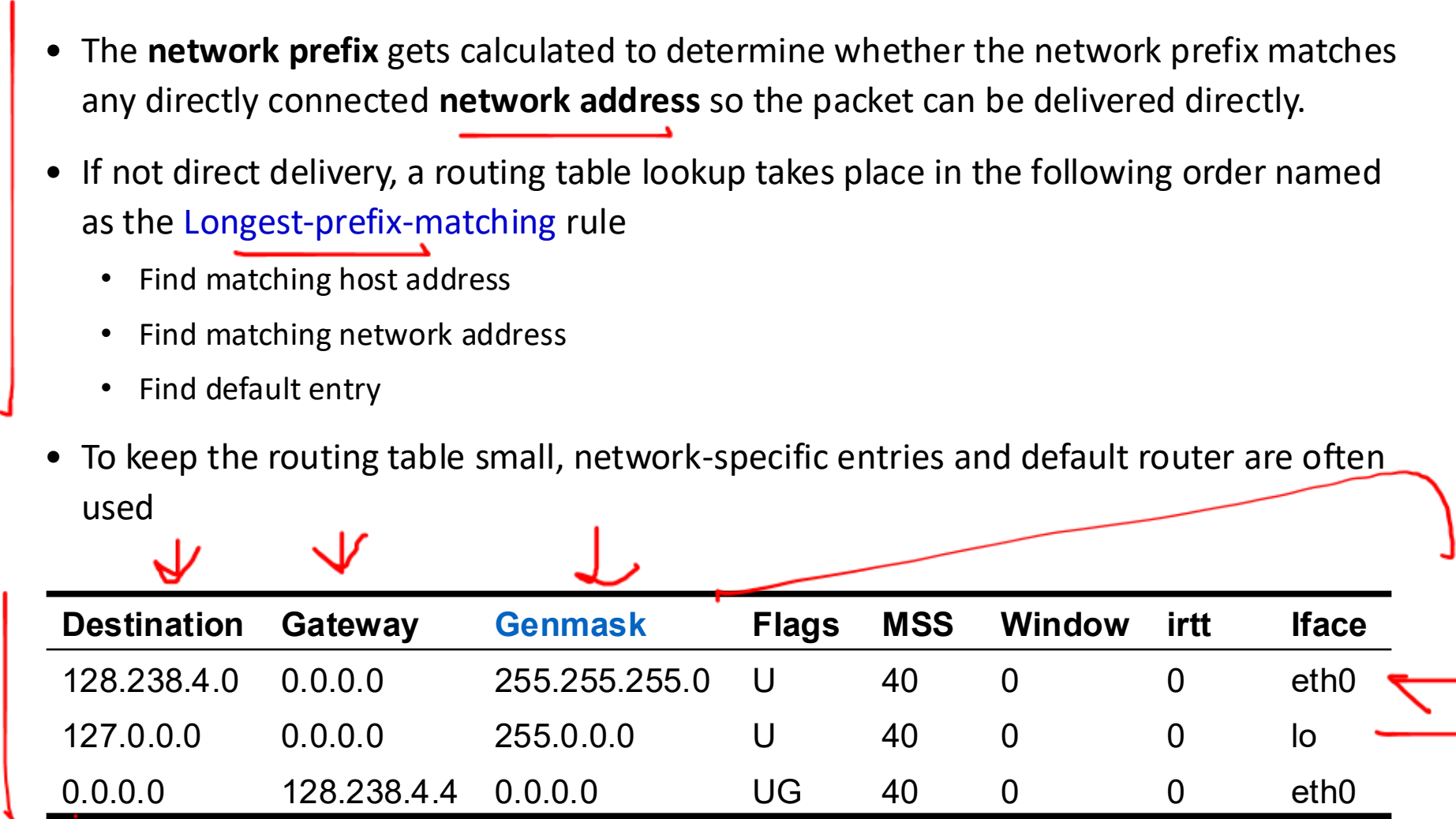


Routing/forwarding table at router R2

<i>Destination</i>	<i>Mask</i>	<i>Next Hop</i>
30.0.0.0	255.0.0.0	40.0.0.7 R1
40.0.0.0	255.0.0.0	Deliver direct
128.1.0.0	255.255.0.0	Deliver direct
192.4.10.0	<u>255.255.255.0</u>	128.1.0.9 ←

Routing Table Lookup - Summary

- To route each IP packet, the destination IP address is first extracted and then
 - The **network prefix** gets calculated to determine whether the network prefix matches any directly connected **network address** so the packet can be delivered directly.
 - If not direct delivery, a routing table lookup takes place in the following order named as the **Longest-prefix-matching** rule
 - Find matching host address
 - Find matching network address
 - Find default entry
- To keep the routing table small, network-specific entries and default router are often used



Destination	Gateway	Genmask	Flags	MSS	Window	irtt	Iface
128.238.4.0	0.0.0.0	255.255.255.0	U	40	0	0	eth0
127.0.0.0	0.0.0.0	255.0.0.0	U	40	0	0	lo
0.0.0.0	128.238.4.4	0.0.0.0	UG	40	0	0	eth0

Getting a datagram from source to destination (1/4)

IP datagram:

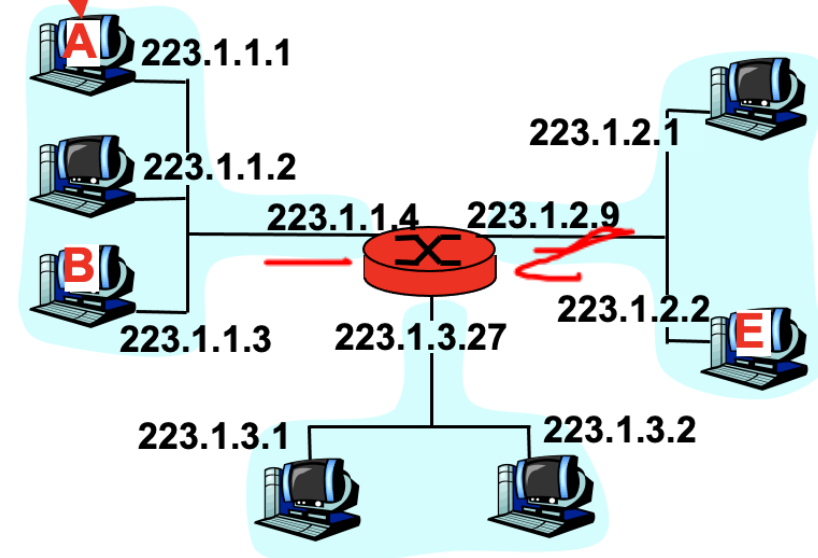
misc fields	source IP addr	dest IP addr	data
-------------	----------------	--------------	------

datagram remains unchanged, as it travels source to destination
addr fields of interest here

routing table in A

Dest. Net.	Next Hop
223.1.1	Deliver direct
223.1.2	223.1.1.4
223.1.3	223.1.1.4

124



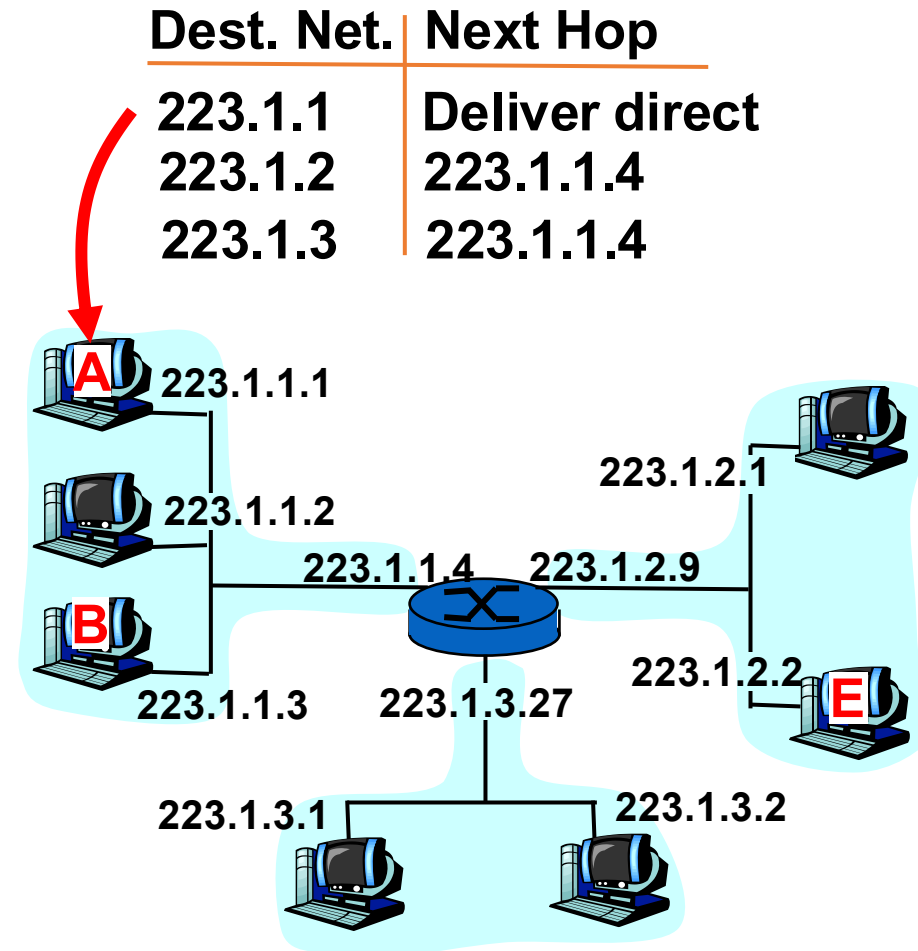
Getting a datagram from source to destination (2/4)

misc fields	223.1.1.1	223.1.1.3	data
-------------	-----------	-----------	------

Starting at A, given IP datagram addressed to B:

look up net. address of B
find B is on same net. as A
link layer will send datagram directly to B inside link-layer frame

B and A are directly connected



Getting a datagram from source to destination (3/4)

misc fields	223.1.1.1	<u>223.1.2.2</u>	data
-------------	-----------	------------------	------

Starting at A, dest. E:

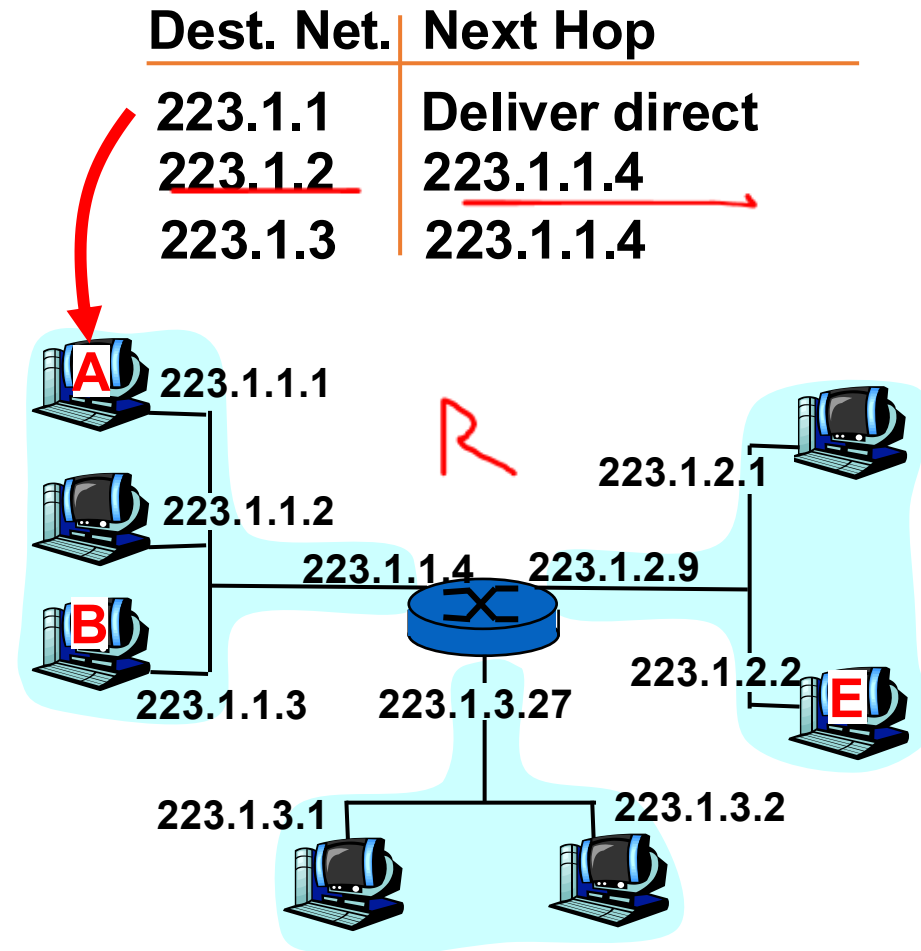
look up network address of E
E on *different* network

A, E not directly
attached

routing table: next hop router to
E is 223.1.1.4

link layer sends datagram to
router 223.1.1.4 inside link-layer
frame

datagram arrives at 223.1.1.4
continued.....



Getting a datagram from source to destination (4/4)

misc fields	223.1.1.1	223.1.2.2	data
----------------	-----------	-----------	------

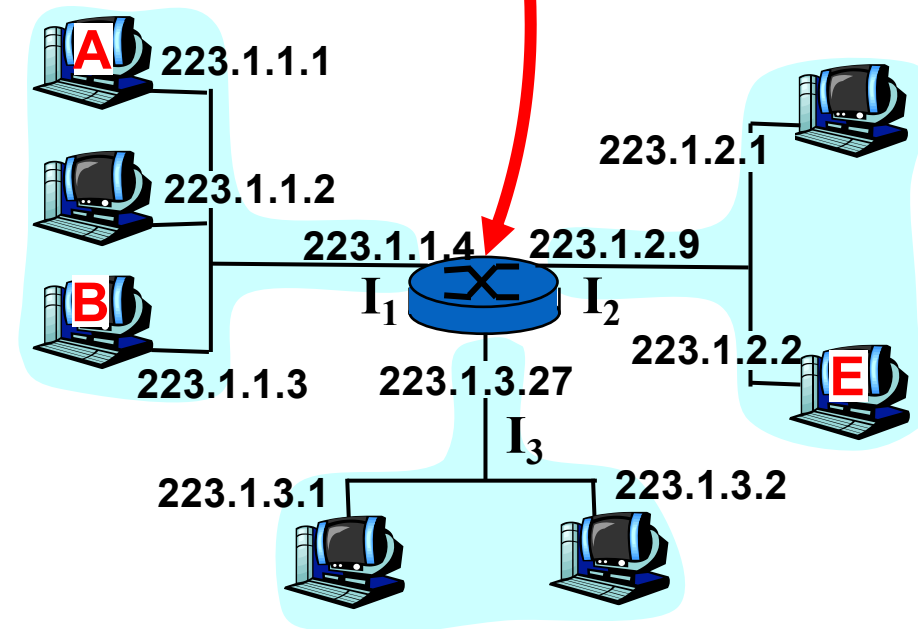
Arriving at 223.1.4,
destined for 223.1.2.2

look up network address of E
E on *same* network as router's
interface 223.1.2.9

router, E directly
attached

link layer sends datagram to
223.1.2.2 inside link-layer frame
via interface I_2 (223.1.2.9)
datagram arrives at 223.1.2.2!!!
(hooray!)

Dest. Net.	Next Hop	Interface
223.1.1	Deliver direct	I_1
223.1.2	Deliver direct	I_2
223.1.3	Deliver direct	I_3



IP addressing: last words ...

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers
<http://www.icann.org/>

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)

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IPv6: Motivation

- IPv4 limitations

- Shortage of address space
 - Subnetting, CIDR, dynamic resource allocation through DHCP, NAT only temporary solutions
- Need better support for real-time audio/video
 - No support for resource allocation/ reservation
- No encryption or authentication

- IPv6 addresses all the above IPv4 limitations

- Also has better header format and newer options

IPv6: Additional Advantages

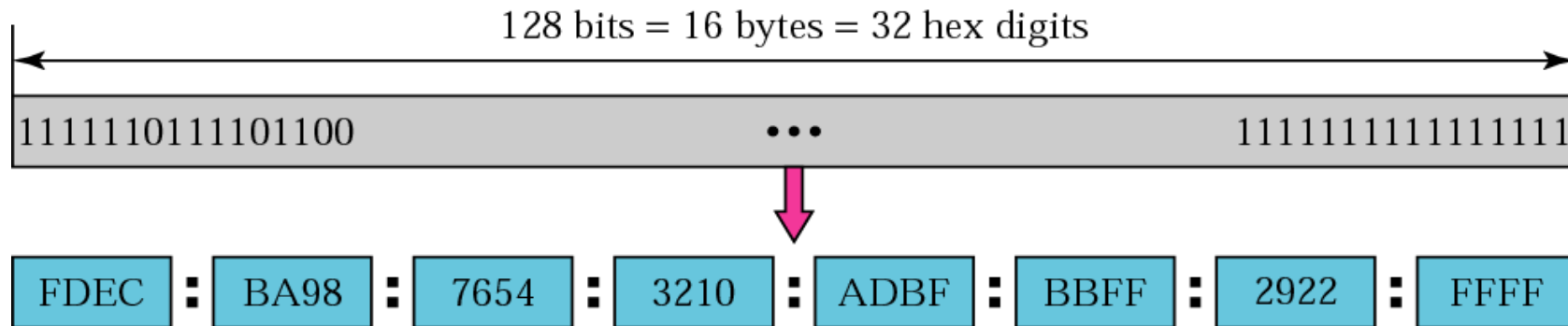
- Streamlined 40-byte header
- Hierarchical addressing
- Updated fragmentation and reassembly procedures
- Improved options
- Provision for protocol extension, transition capabilities, tunneling
- Support for autoconfiguration and renumbering, without need for DHCP

IPv6 addresses

32

2^6

- 128 bits
- Usually expressed in *hexadecimal colon notation*



[src: Forouzan]

IPv6 address abbreviation

- *Leading* 0s of a section (four hex digits between two colons) can be omitted
- *Any one* run of 0 sections can be abbreviated

Unabbreviated

FDEC : BA98 : 0074 : 3210 : 000F : BBFF : 0000 : FFFF



FDEC : BA98 : 74 : 3210 : F : BBFF : 0 : FFFF

Abbreviated

Abbreviated

FDEC : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF

[src: Forouzan]

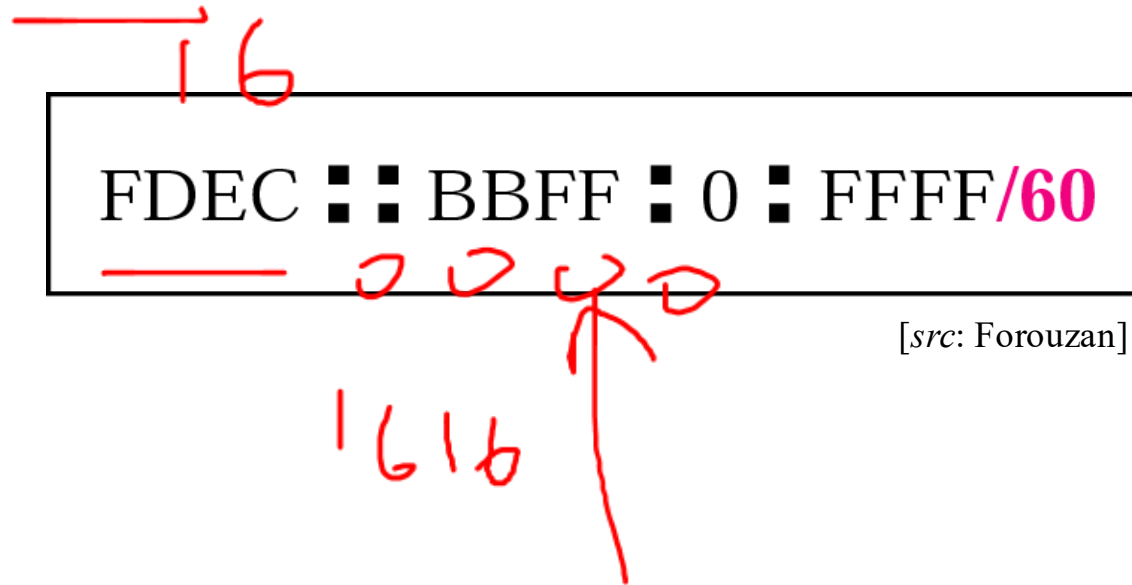


FDEC :: BBFF : 0 : FFFF

More Abbreviated

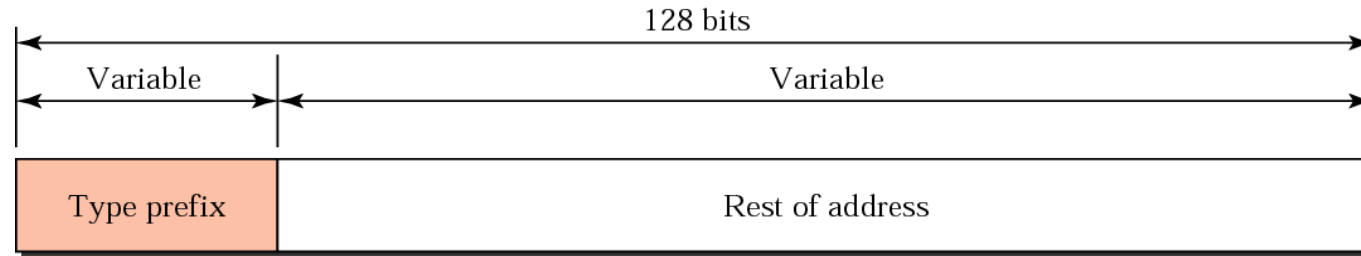
IPv6 addresses: CIDR notation

- /x notation used, as in IPv4



[src: Forouzan]

Address Structure



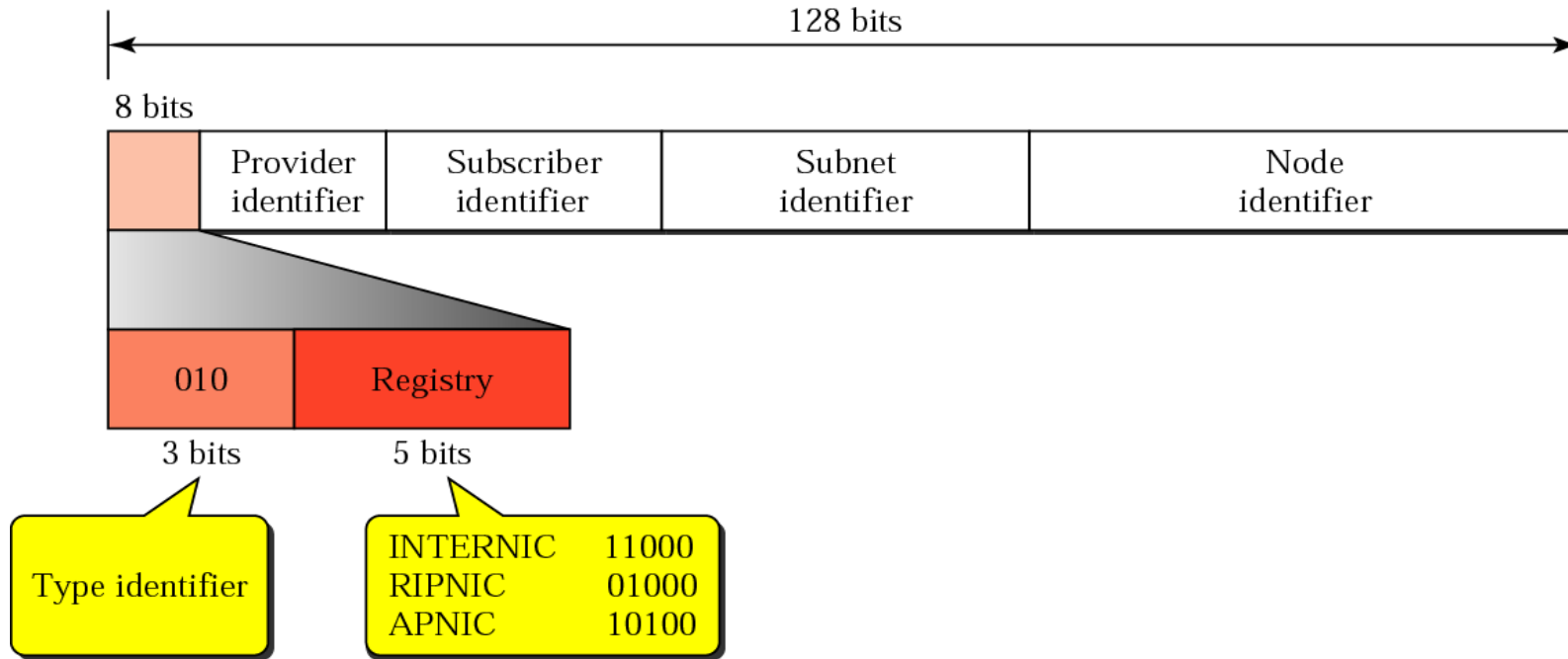
[src: Forouzan]

Type Prefix	Type	Fraction
010	Provider-based unicast addresses	1/8
011	Reserved	1/8
100	Geographic unicast addresses	1/8
101	Reserved	1/8
110	Reserved	1/8
1110	Reserved	1/16
1111 0	Reserved	1/32
1111 10	Reserved	1/64
1111 110	Reserved	1/128
1111 1110 0	Reserved	1/512
1111 1110 10	Link local addresses	1/1024
1111 1110 11	Site local addresses	1/1024
1111 1111	Multicast addresses	1/256

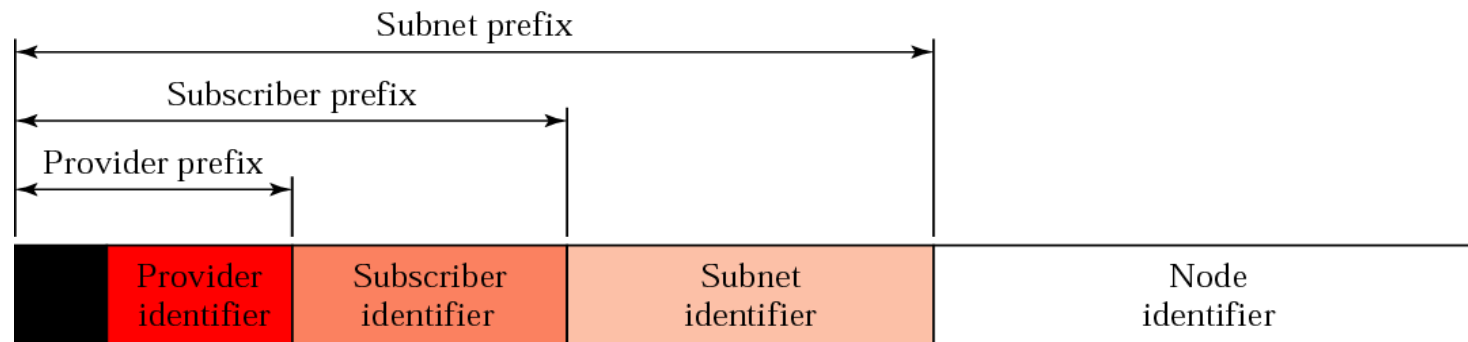
The rest of the addresses is mostly *reserved* or *unassigned*

[src: Forouzan]

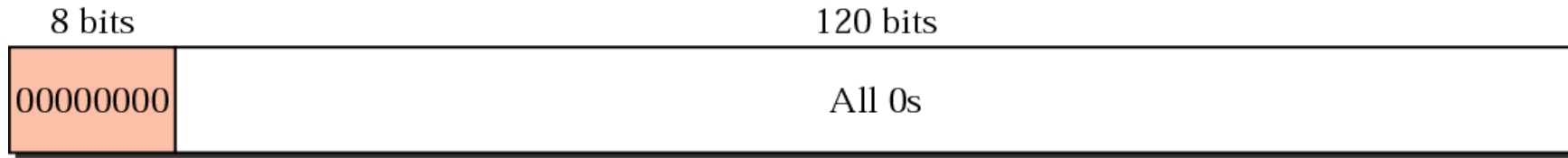
Provider based addresses



[src: Forouzan]

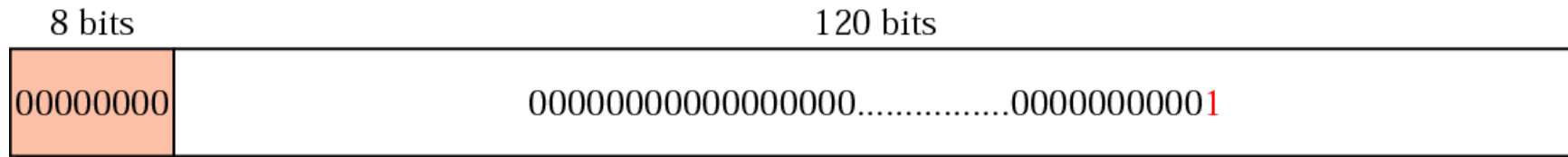


Unspecified and loopback addresses



Unspecified address

[src: Forouzan]

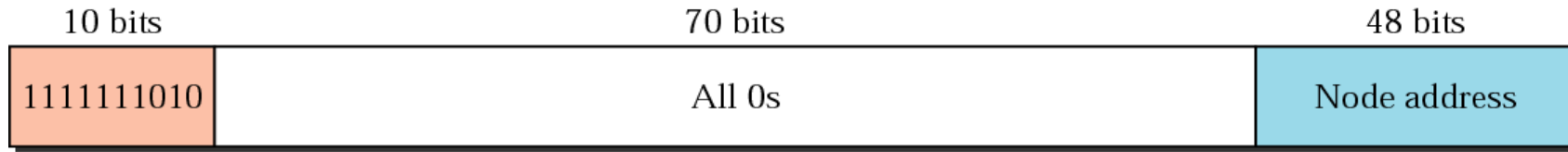


Loopback address

[src: Forouzan]

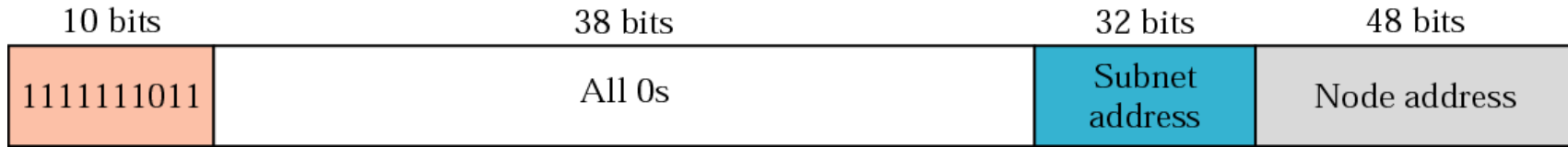
127.0.0.1

Local addresses



Link local address

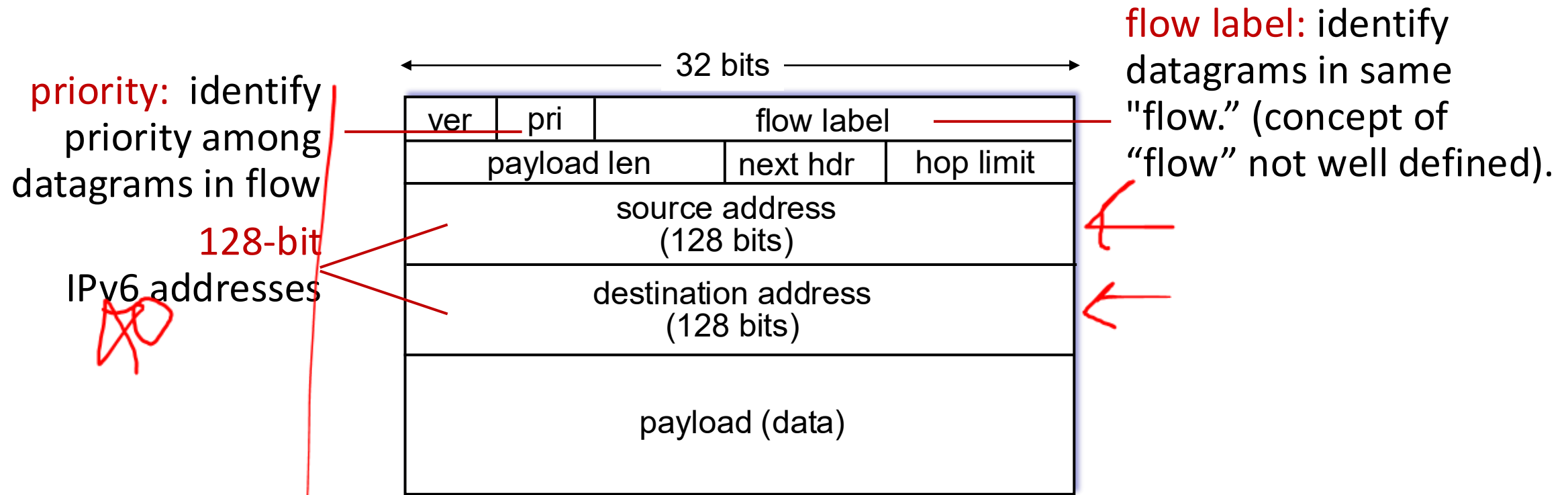
[src: Forouzan]



Site local address

[src: Forouzan]

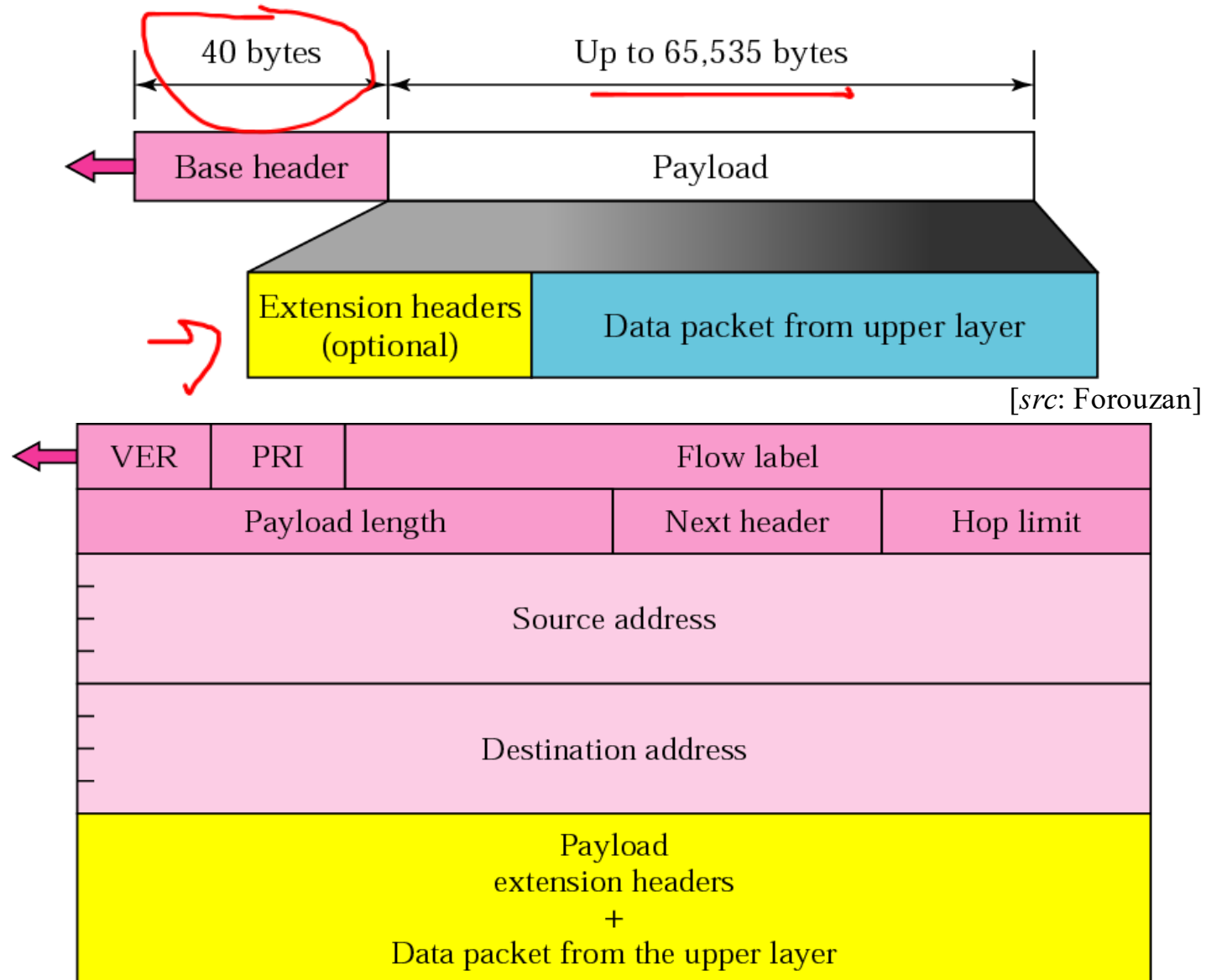
IPv6 datagram format



What's missing (compared with IPv4):

- no checksum (to speed processing at routers)
- no fragmentation/reassembly
- no options (available as upper-layer, next-header protocol at router)

IPv6 datagram format



IPv6 vs IPv4: Header format comparison

<i>Comparison</i>	
1.	The header length field is eliminated in IPv6 because the length of the header is fixed in this version.
2.	The service type field is eliminated in IPv6. The priority and flow label fields together take over the function of the service type field.
3.	The total length field is eliminated in IPv6 and replaced by the payload length field.
4.	The identification, flag, and offset fields are eliminated from the base header in IPv6. They are included in the fragmentation extension header.
5.	The TTL field is called hop limit in IPv6.
6.	The protocol field is replaced by the next header field.
7.	The header checksum is eliminated because the checksum is provided by upper layer protocols; it is therefore not needed at this level.
8.	The option fields in IPv4 are implemented as extension headers in IPv6.

[src: Forouzan]



Priority / Traffic class field values

<i>Priority</i>	<i>Meaning</i>
0	No specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

For congestion-
controlled traffic

[src: Forouzan]

<i>Priority</i>	<i>Meaning</i>
8	Data with most redundancy
.	.
.	.
.	.
15	Data with least redundancy

For non congestion-
controlled traffic

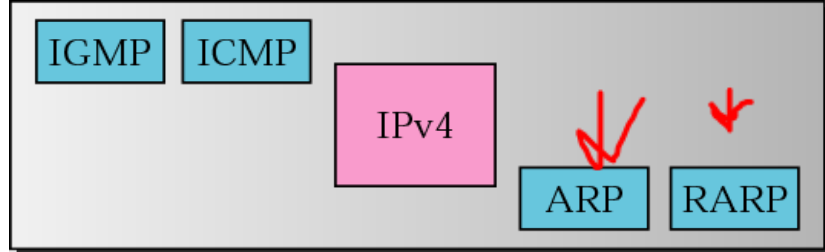
Next header codes

<i>Code</i>	<i>Next Header</i>
0	Hop-by-hop option
2	ICMP
6	TCP
17	UDP
43	Source routing
44	Fragmentation
50	Encrypted security payload
51	Authentication
59	Null (No next header)
60	Destination option

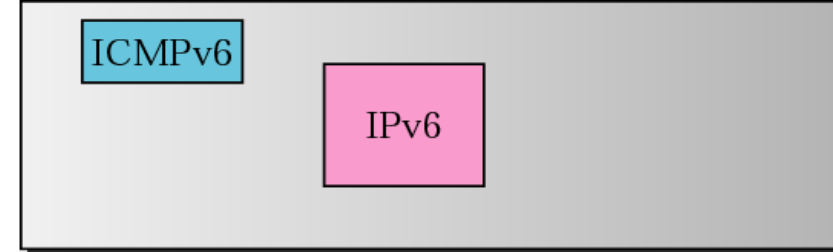
[src: Forouzan]

IPv4 vs IPv6 network layers

- RARP dropped (BOOTP/DHCP used instead)
- ICMPv6 combines IGMP and ARP functionality



Network layer in version 4

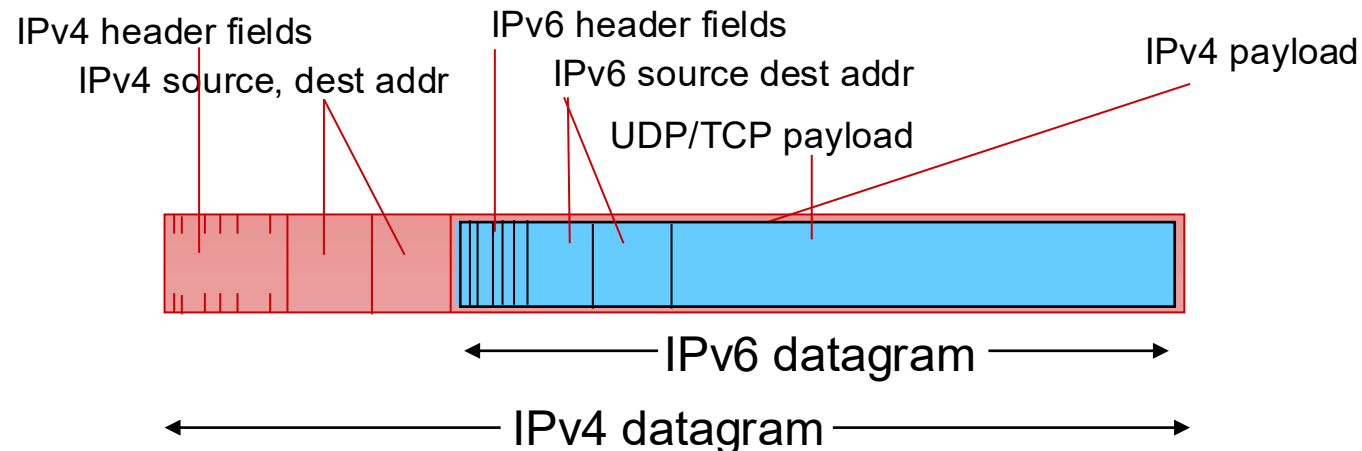


Network layer in version 6

[src: Forouzan]

Transition from IPv4 to IPv6

- not all routers can be upgraded simultaneously
 - no “flag days”
 - how will network operate with mixed IPv4 and IPv6 routers?
- **tunneling**: IPv6 datagram carried as *payload* in IPv4 datagram among IPv4 routers (“packet within a packet”)
 - tunneling used extensively in other contexts (4G/5G)

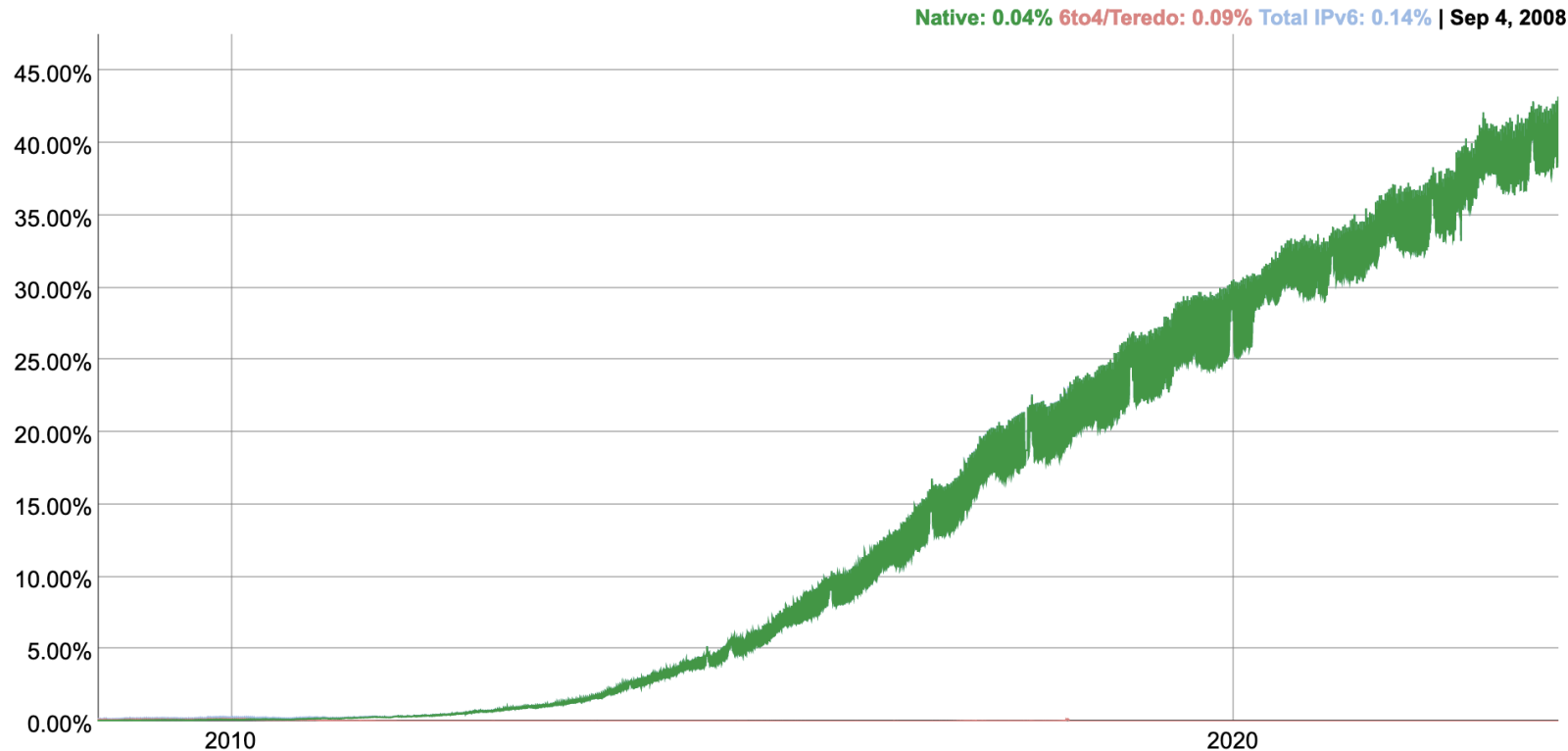


IPv6: adoption

- Google¹: ~ 40% of clients access services via IPv6 (2023)
- NIST: 1/3 of all US government domains are IPv6 capable

IPv6 Adoption

We are continuously measuring the availability of IPv6 connectivity among Google users. The graph shows the percentage of users that access Google over IPv6.



IPv6: adoption

- Google¹: ~ 40% of clients access services via IPv6 (2023)
- NIST: 1/3 of all US government domains are IPv6 capable
- Long (long!) time for deployment, use
 - 25 years and counting!
 - think of application-level changes in last 25 years: WWW, social media, streaming media, gaming, telepresence, ...
 - *Why?*

¹ <https://www.google.com/intl/en/ipv6/statistics.html>